

EDUCATION

Cornell University

Ph.D. in Astronomy and Space Sciences.

Ithaca, NY

2021–Present

- Thesis: “The Epoch of Reionization Spectrometer” (in preparation). Expected completion: Spring 2027.

Pontifical Catholic University of Peru

B.Sc. in Electronics Engineering. ABET accredited.

Lima, Peru

2011–2016

- Thesis: “Design of a LabVIEW-based polyphase filter bank spectrometer for radio astronomy using FlexRIO FPGA technology and CUDA-enabled GPU”.

EXPERIENCE

CCAT collaboration

Graduate Research Assistant. Advisors: Prof. Gordon Stacey and Prof. Michael Niemack

Ithaca, NY

2021 –Present

- Design, fabrication and deployment of state of the art sub-mm instrumentation for the CCAT collaboration.
- Opto-mechanical design, cryogenic systems, RF engineering, superconducting detectors, and digital readout electronics.

Atacama Cosmology Telescope

Site Engineer. Supervisors: Prof. Suzanne Staggs and Prof. Mark Devlin.

San Pedro de Atacama, Chile

Oct 2019 –Jul 2021

- Ensured the continuous and smooth operation of all of the telescope’s subsystems, such as a the cryogenic cooling system $\sim 90\text{mK}$).
- Performed calibration procedures like the alignment of the receiver and the primary and secondary reflectors through high-precision photogrammetry ($\sim 10\mu\text{m}$). Carried out frequency response measurements of the detector arrays through Fourier Transform Spectroscopy.
- Participated in the deployment of the Advance ACTPol low-frequency detector array (27/39 GHz). Assisted in the installation of the new chain of filters and cold silicon reimaging optics.

Management Solutions

Assistant Business Consultant

Lima, Peru

Sep 2018 –Jul 2019

- Worked in big data and bank stress test for a major local bank in Lima, Peru. Software development and data science for a leading financial institution in Madrid, Spain.

Institute for Radioastronomy

Research Engineer. Supervisor: Prof. Jorge Heraud

Lima, Peru

May 2017 –Aug 2018

- Worked in the design and construction of an 8-meter diameter radio telescope (RT8). Designed several hardware and software systems for the telescope’s electromechanical pointing system.
- Assisted in the characterization of an hydrogen 21-cm line feedhorn and its RF over Fiber (RFoF) frontend. Developed an antenna test system based on a software-defined radio technology.

PUBLICATIONS

- [1] B. Keller, **R. Freundt**, J. R. Burgoyne, S. Chapman, S. Choi, C. J. Duell, C. Groppi, C. Humphreys, L. T. Lin, A. Middleton, M. D. Niemack, D. Patel, E. Vavagiakis, S. Walker, Y. Wang, and R. Xie, “Ccat: Flexible stripline circuits for large-format kinetic inductance detector (kid) array readout”, *IEEE Transactions on Applied Superconductivity*, vol. 36, no. 6, pp. 1–6, 2026.
- [2] **R. Freundt**, Y. Li, D. Henke, J. Austermann, J. R. Burgoyne, S. Chapman, S. K. Choi, C. J. Duell, Z. Huber, M. Niemack, T. Nikola, L. Lin, D. A. Riechers, G. Stacey, A. K. Vaskuri, E. M. Vavagiakis, J. Wheeler, and B. Zou, “CCAT: A status update on the EoR-Spec instrument module for Prime-Cam”, in *Millimeter, Submillimeter, and Far-Infrared Detectors and Instrumentation for Astronomy XII*, vol. 13102, SPIE, Aug. 16, 2024, pp. 343–352.
- [3] C.-P. Collaboration, M. Aravena, J. E. Austermann, *et al.*, “CCAT-prime Collaboration: Science Goals and Forecasts with Prime-Cam on the Fred Young Submillimeter Telescope”, *The Astrophysical Journal Supplement Series*, vol. 264, no. 1, p. 7, Dec. 2022, ISSN: 0067-0049.
- [4] **R. Freundt** and J. A. Heraud, “Design of a LabVIEW-based polyphase filter bank spectrometer for radio astronomy using FlexRIO FPGA technology and CUDA-enabled GPU”, in *2017 XXXIInd General Assembly and Scientific Symposium of the International Union of Radio Science (URSI GASS)*, Aug. 2017, pp. 1–4.

TEACHING

- **Teaching Assistant** at Cornell University Fall 2022, 2024, 2025
Multiwavelength astronomical techniques (ASTRO 4410)
- **Teaching Assistant** at Cornell University Spring 2023
Our Solar System (ASTRO 1102)
- **Teaching Assistant** at PUCP Aug 2017 - Dec 2017
Space Science and Engineering (ING306)

SKILLS

- **Programming:** Python, C/C++, CUDA, LabVIEW
- **CAD:** Solidworks, KiCAD
- **Tools:** LaTeX, Git, Docker, Linux
- **Other:** Ansys Zemax Optics Studio, Wilderness Advanced First Aid (WAFA) certification

SCHOLARSHIPS AND AWARDS

- Beca de Estímulo Académico Solidario (BEAS) 2013–2016
Fully-funded undergraduate scholarship for outstanding academic performance and financial need.
- URSI Young Scientist Award Aug 2017
Merit-based registration fee waiver and travel grant to attend URSI 2017, Montreal, QC.
- Summer School of Astronomical Instrumentation, Dunlap Institute, UofT Jul 2019
Merit-based tuition waiver and travel grant. Toronto, ON.
- CASPER Workshop, Harvard & Smithsonian Center for Astrophysics (CfA) Aug 2019
Merit-based fee waiver and travel grant (invited, could not attend). Cambridge, MA.